

# 22AIE213 – Machine Learning

## Course Project Guidelines

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### 1 Project Objectives

One of the course requirements is to complete a project in a group of two or three students. The course project provides an opportunity to explore an interesting machine learning problem of your choice using real-world data.

The project needs the following components:

- Study the main ideas from a few machine learning research papers on the problem domain
- Conduct experiments by selecting a real-world dataset and applying appropriate machine learning algorithms.
- Modification or extension of the baseline algorithm to improve performance
- Justification for model choices and modifications
- Critical analysis of results and limitations
- Work on a theoretical question in machine learning and relate it to the problem.

As part of the project, students are expected to select a real-world dataset, apply at least one existing machine learning algorithm, and modify or extend the baseline approach to improve performance. All model choices and modifications should be theoretically justified, and the project should include a critical analysis of the results, assumptions, and limitations of the proposed approach.

### 2 Sample Project Domains

Students may choose a project from (**but not limited to**) the following domains:

- |                                                          |                                                           |
|----------------------------------------------------------|-----------------------------------------------------------|
| • Climate change analysis and carbon emission prediction | systems                                                   |
| • Renewable energy forecasting and optimization          | • Smart cities and resource optimization                  |
| • Air and water quality monitoring                       | • Disease risk prediction and diagnosis support           |
| • Sustainable agriculture and crop yield prediction      | • Patient outcome and survival analysis                   |
| • Poverty and socio-economic indicator analysis          | • Healthcare resource allocation                          |
| • Disaster prediction and early warning                  | • Epidemic and outbreak prediction                        |
|                                                          | • Personalized medicine and treatment response prediction |

- Predictive maintenance for industrial equipment
- Quality inspection and defect detection
- Supply chain demand forecasting
- Process optimization in manufacturing
- Energy consumption analysis in industries
- Fault detection and anomaly detection
- Gene expression analysis and classification
- DNA/RNA sequence analysis
- Protein structure and function prediction
- Cell type classification from single-cell data
- Population genetics analysis
- Evolutionary pattern detection
- Drug–target interaction prediction
- Virtual screening and molecular docking prediction
- Drug repurposing using ML
- Climate and weather forecasting
- Financial time-series analysis
- Healthcare monitoring signals
- Industrial sensor data analysis
- Energy demand and load forecasting
- Fairness and bias analysis in ML systems
- Education analytics and student performance prediction
- Public policy impact assessment
- Urban mobility and traffic prediction
- Social network analysis
- Algorithmic trading
- Credit risk modeling
- Market prediction
- Student engagement modeling
- Adaptive learning systems
- Personalization systems
- ML on resource-constrained devices
- Privacy-preserving ML
- Human-in-the-loop ML

**Important:** Students must proceed with a project *only if* a publicly available dataset exists for the chosen topic. Projects without accessible public datasets will not be approved.

### 3 Public Dataset Sources

Students are encouraged to use publicly available datasets from the following sources:

- **Kaggle Datasets** — Thousands of datasets across multiple domains.  
<https://www.kaggle.com/datasets>
- **UCI Machine Learning Repository** — A classic and widely used collection of machine learning datasets.  
<https://archive.ics.uci.edu/ml/index.php>
- **Google Dataset Search** — A search engine for discovering datasets across the web.  
<https://datasetsearch.research.google.com>
- **AWS Open Data Registry** — Large-scale public datasets hosted on AWS.  
<https://registry.opendata.aws>
- **Microsoft Research Open Data** — Curated datasets released by Microsoft Research.  
<https://msropendata.com>

## 4 Recommended Machine Learning Project Pipeline

Students are expected to follow a structured machine learning workflow while working on their projects. A suggested end-to-end pipeline is outlined below (adapted from standard ML best practices):

1. **Ideation and Problem Definition:** Clearly define the problem statement, objectives, and assumptions of the machine learning task.
2. **Data Acquisition and Labeling:** Acquire a suitable publicly available dataset or perform labeling if required.
3. **Exploratory Data Analysis (EDA):** Explore and analyze the data to understand its characteristics, distributions, correlations, and potential issues such as missing values or noise.
4. **Baseline Model Development:** Build a training pipeline for an initial baseline model using an appropriate machine learning algorithm.
5. **Model Evaluation and Error Analysis:** Test the model on unseen data and perform error analysis to identify failure modes.
6. **Model Improvement:** Use insights from error analysis to modify, extend, or improve the model (e.g., feature engineering, regularization, model selection).
7. **Iterative Refinement:** Repeat steps 4–6 until satisfactory model performance is achieved.
8. **Deployment Considerations:** Design the infrastructure required to deploy the model, considering runtime, scalability, and user constraints (conceptual discussion is sufficient; full deployment is not mandatory).
9. **Monitoring and Maintenance:** Discuss how model performance would be monitored over time and how feedback could trigger retraining or updates.

*Reference:* <https://www.mihaileric.com/posts/setting-up-a-machine-learning-project/>

## 5 Course Project Description

One of the main objectives of this course is to prepare students to apply machine learning algorithms to real-world problems. The course project provides an opportunity to explore a machine learning problem of your choice using real data, either empirically or theoretically.

Through this project, students are expected to:

- Apply existing machine learning algorithms to a real-world dataset.
- Analyze model performance and identify failure modes.
- Modify or extend baseline algorithms to improve performance.
- Provide theoretical justification for model choices and improvements.

## 6 Sample Project Context

As an illustrative example, consider a project focuses on improving personalized learning in online education platforms such as Khan Academy and Coursera. Since it is difficult to directly measure students' understanding in online settings, these platforms use diagnostic multiple-choice questions designed to reveal common misconceptions through incorrect answers.

The goal of the project is to build machine learning models that predict whether a student will correctly answer a given diagnostic question based on their previous responses and the responses of other students. Such predictions help estimate a student's ability level and enable personalized features such as recommending questions of appropriate difficulty. The project involves applying existing machine learning algorithms, comparing their performance, analyzing their strengths and weaknesses, and then modifying or extending these algorithms to improve prediction accuracy.

The project is divided into three main parts: **Part A: Baseline Modeling** , **Part B: Model Improvement** and **Part C: Solving a small theoretical problem in ML related to the domain**. Students are encouraged to clearly separate these parts in their report.

### 6.1 Part A: Baseline Modeling

In this part, students will establish strong baseline models and analyze their performance.

1. **Dataset Selection and Problem Formulation:** Select a publicly available real-world dataset and clearly define the machine learning task (e.g., classification, regression, recommendation). Specify the input features, output targets, and assumptions made.
2. **Data Preprocessing and Exploration:** Perform exploratory data analysis to understand data characteristics such as distributions, missing values, noise, and correlations. Justify any preprocessing steps including normalization, feature selection, or data splitting.
3. **Baseline Model Implementation:** Implement at least one baseline machine learning model appropriate for the task. Baseline models should use standard algorithms covered in the course (e.g., linear models, k-nearest neighbors, decision trees, or neural networks).
4. **Model Training and Evaluation:** Train the baseline model(s) using appropriate evaluation metrics. Clearly describe the training procedure, validation strategy, and hyperparameter selection.
5. **Error Analysis:** Analyze the failure modes of the baseline model. Identify patterns where the model performs poorly and discuss possible reasons based on data properties or model assumptions.

### 6.2 Part B: Model Improvement

In this part, students will build upon the baseline models to improve performance or address identified limitations.

1. **Motivation for Improvement:** Based on the error analysis from Part A, identify specific weaknesses of the baseline model that motivate further improvement.
2. **Proposed Modification or Extension:** Propose modification or extension to the baseline model. This may include feature engineering, regularization, architectural changes, alternative loss functions, or algorithmic enhancements.

3. **Theoretical Justification:** Provide a theoretical explanation for why the proposed modification is expected to improve model performance or robustness.
4. **Implementation and Evaluation:** Implement the modified model and evaluate it using the same metrics and validation strategy as the baseline to ensure a fair comparison.
5. **Comparative Analysis:** Compare the improved model with baseline results and discuss observed gains or trade-offs, including any new limitations introduced.

### 6.3 Part C: Solving a small theoretical problem

In this part, students are expected to investigate and solve a small but well-defined theoretical problem in machine learning that is related to the chosen application domain. Select a focused theoretical question relevant to the problem domain or learning task. Examples include questions related to generalization, model complexity, bias–variance tradeoff, regularization, optimization, loss functions, or sample complexity.

**Sample Problem Description and Theoretical Questions** Carbon emission prediction is a challenging real-world machine learning problem that involves modeling complex relationships between emissions and factors such as energy consumption, industrial activity, transportation, population growth, and policy interventions. The data are often noisy, heterogeneous across regions and sectors, and limited in historical coverage. As a result, choosing an appropriate model complexity is critical to achieving good generalization while maintaining interpretability and robustness, especially in policy-oriented applications.

Students may address one or more of the following theoretical questions related to model complexity in the context of carbon emission prediction:

1. How does increasing model complexity affect generalization performance when predicting carbon emissions across different regions or economic sectors?
2. How do underfitting and overfitting arise in carbon emission prediction models, and what theoretical indicators can be used to identify each case?
3. Given limited historical carbon emission data, how should model complexity be selected using principles from learning theory?
4. How does model complexity influence the bias–variance tradeoff for short-term versus long-term carbon emission forecasts?
5. Why are low-complexity and interpretable models often preferred in carbon emission policy analysis, even when more complex models achieve higher predictive accuracy?
6. How does dataset size constrain the choice of model complexity in carbon emission prediction tasks?
7. In what ways does regularization act as a mechanism to control model complexity and improve generalization in emission prediction models?
8. How does adding more features (e.g., energy sources, industrial indicators, transportation metrics) increase effective model complexity, even for simple models?
9. How does modeling temporal dependencies, such as lag variables and trends, increase model complexity, and what theoretical risks does this introduce?
10. How does tree depth affect model capacity and generalization in tree-based models for carbon emission regression?

11. How do depth and width influence the expressive power and complexity of neural networks used for emission prediction?
12. Why might highly complex, overparameterized models still generalize well in carbon emission prediction despite classical learning theory expectations?
13. How do theoretical model selection principles, such as the bias–variance tradeoff and Occam’s razor, apply to carbon emission prediction problems?
14. Why are simpler models often more robust to noise and measurement errors in carbon emission data compared to complex models?
15. Is there a theoretical limit beyond which increasing model complexity no longer improves carbon emission prediction performance? Explain why.

## 7 Expected Deliverables

- **Project Proposal** (1–2 pages)
- **Literature Review Document** (4–5 pages)
- **Final Report** (8–10 pages)
- **Code** with proper documentation
- **Presentation** (10–12 slides)

## 8 Submission Instructions

All project materials must be submitted through the AUMS by the specified deadline.